

Def.

Let $E \subset X$ be a non-empty subset of a Metric space X . Define

$$P_E: X \rightarrow \mathbb{R}; x \mapsto \inf_{z \in E} d(x, z)$$

is called a distance from x to E .

Prop. Let P_E be a distance from x to E in a Metric space X .

1). $P_E(x) = 0$ if and only if $x \in \overline{E}$.

2). P_E is Continuous Uniformly.

Proof. To show the first, using the property of closure and infimum

$$x \in \overline{E} \Leftrightarrow \forall \varepsilon > 0, B_\varepsilon(x) \cap E \neq \emptyset \Leftrightarrow \forall \varepsilon > 0, \exists y \in E \text{ such that } d(x, y) < \varepsilon \Leftrightarrow \inf_{z \in E} d(x, z) = 0.$$

To show the second assertion, observe that: given $x, y \in E$,

$$P_E(x) \leq d(x, z) \leq d(x, y) + d(y, z) \text{ for any } z \in E$$

Since it holds for all $z \in E$,

$$P_E(x) \leq d(x, y) + P_E(y)$$

and which implies $P_E(x) - P_E(y) \leq d(x, y)$. Similarly, $P_E(y) - P_E(x) \leq d(x, y)$, so

$$|P_E(x) - P_E(y)| \leq d(x, y) \text{ for any } x, y \in X.$$

Given $\varepsilon > 0$, take $\delta = \varepsilon$, then we obtain that

$$d(x, y) < \delta \Rightarrow |P_E(x) - P_E(y)| \leq d(x, y) < \varepsilon$$

□

22. Let $A, B \subseteq X$ be non-empty disjoint closed sets in metric space X .

Show that a map

$$f: X \rightarrow \mathbb{R}; x \mapsto \frac{p_A(x)}{p_A(x) + p_B(x)}$$

is continuous for $y=0$ iff $x \in A$ and $f(x)=1$ iff $x \in B$.

Proof Since there is no element in X s.t.

$$p_A(x) + p_B(x) = 0$$

because it is equivalent to $p_A(x) = p_B(x) = 0$, but it implies that

$$x \in \overline{A \cap B} = A \cap B = \emptyset$$

by assumption, which is contradiction. Therefore this f is well-defined.

The continuity is clear. And, observe that

$$\frac{p_A(x)}{p_A(x) + p_B(x)} = 0 \text{ iff } p_A(x) = 0 \text{ iff } x \in A = \overline{A}.$$

and

$$\frac{p_B(x)}{p_A(x) + p_B(x)} = 1 \text{ iff } p_B(x) = 0 \text{ iff } x \in B = \overline{B}.$$

Every closed subset $A \subseteq X$ is $Z(f)$ for some continuous map $f: X \rightarrow \mathbb{R}$.

Proof. If $A = X$, then put $f=0$. Suppose that $A \subsetneq X$.

Then, there exists non-empty closed set $B \subset X$ s.t. $A \cap B = \emptyset$. (For example, $B = \{b\}$, $b \notin A$)

Now, $f(x) = \frac{p_A(x)}{p_A(x) + p_B(x)}$, then $Z(f) = A$.

Moreover, observe that

$$A = f^{-1}[\{0\}] \subseteq f^{-1}[[0, \frac{1}{2}]] \text{ and } B = f^{-1}[\{1\}] \subseteq f^{-1}[[\frac{1}{2}, 1]]$$

implies that A and B are covered by disjoint open sets.